## 15.2 Semester II

| Code: BCA-2001 T | CC-III | Mathematical Foundation to Computer Science – II | 3L+T: 0P | 3 Credits (45 hours theory) |
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**Max Marks: 100; Theory: 100 (Int: 25; Ext: 75)**

**Course Outcomes:** Upon completion of the course, students will be able to

**CO1:** understand correct lines of arguments and proofs.

**CO2:** Understand mathematical techniques that are foundations for understanding advanced computational methods, including numerical methods and optimization.

**CO3:** understand various problem-solving strategies and methods to tackle both theoretical and practical challenges in computer science.

| Unit | Topics | Purposed lectures |
|---|---|---|
| I | **Logic and Methods of Proofs**<br>Propositions, logical operations (basic connectives), compound statements, construction of truth table, quantifiers, conditional statements, tautology, contradiction, logical equivalence. Conjunctive Normal Forms (CNF) and Disjunctive Normal Forms (DNF).<br>**Methods of Proofs:** Rules of inference for propositional logic, modus ponens, modus tollens, syllogism, proof by contradiction, Mathematical Induction. | 11 |
| II | **Algebraic Structures**<br>Semi-group, Monoid, Group, Abelian Group, Subgroup, Properties of Subgroup, Cyclic group. | 11 |
| III | **Numerical Methods**<br>Concept and importance of errors in numerical methods.<br>**Solution of algebraic and transcendental equations:** Bisection method and Newton-Raphson methods.<br>**Numerical Interpolation:** Newton's Forward and Newton's Backward interpolation formula and Lagrange's formula.<br>**Numerical Integration:** Quadrature Formula, Trapezoidal rule, Simpson's 1/3 rule and Simpson's 3/8 rule (only formulae and applications for all the topics mentioned in this unit) | 11 |
| IV | **Optimization Techniques**<br>**Linear programming:** Introduction, LP formulation, Graphical method for solving LPs with two variables, Special cases in graphical methods, Simplex method, Duality.<br>**Transportation problem:** Definition, Linear form, North-west corner method, least cost method, Vogel's approximation method for finding feasible solution, MODI method for finding optimum solution. | 12 |

**Text Books:**


1. Kolman B., Busby R. and Ross S., *Discrete Mathematical Structures*, 6th Edition, Pearson Education, 2015.
2. Sastry S. S., *Introductory Methods of Numerical Analysis*, Fifth Edition, PHL, 2022.
3. Taha Hamdy A., *Operations Research: An Introduction*, Eighth Edition, Pearson Prentice Hall, 2003.


**Reference Books:**


1. Rosen Kenneth H. and Krithivasan Kamala, *Discrete Mathematics and its Applications*, McGraw Hill, India, 2019.
2. Chakravorty J. G. and Ghosh P. R., *Linear Programming and Game Theory*, Moulik Library, 2017.
3. Sharma J. K., *Operations Research: Theory and Applications*, Fourth Edition, Macmillan Publishers, 2007.
4. S.D. Sharma, *Operations Research (Theory Methods & Applications)*, 2014.
5. Discrete Mathematics and Structures, Satinder Bal Gupta, McGrawHill, 2010.

| Code: BCA-2002T (For Theory) BCA- 2002P (For Practical) | CC-IV | Data Structures | 3L+T: 4P | 5 Credits (45 hours theory and 60 hours practical) |
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**Max Marks; Theory: 100 (Int: 25; Ext: 75); Practical: 100**

**Course Outcomes:** Upon competition of the course, students will be able to

**CO1:** Understand the fundamental concepts of data structures and their classifications, including arrays and their operations.

**CO2:** Develop the ability to implement and manipulate linked lists and apply hashing techniques for efficient data storage and retrieval.

**CO3:** Apply stack, queue, and recursion concepts to solve complex computational problems in c programming

**CO4:** Implement tree and graph data structures to solve real-world problems and optimize data organization.

| Unit | Topics | Purposed lectures |
|---|---|---|
| I | **Definition, Classification and Operations of Data Structures:** Definition of data structures, Types of Data Structures: Linear and Non-Linear Data Structure, Algorithms: Complexity, Time-Space Tradeoff. Difference between algorithm and programs.<br>**Arrays:** Definition and Classification of Arrays, Representation of Linear Arrays in Memory,<br>**Operations on Linear Arrays:** Traversing, Inserting, Deleting, Searching, Sorting and Merging.<br>**Searching:** Linear Search and Binary Search, Comparison of Methods.<br>**Sorting:** Bubble Sort, Selection Sort, and Insertion Sort.<br>**Multi-Dimensional Arrays:** Representation of Two-Dimensional Arrays in Memory, Matrices and Sparse Matrices, Multi-Dimensional Arrays. | 11 |
| II | **Linked Lists and Hashing**<br>**Linked Lists:** Definition, Comparison with Arrays, Representation, Types of Linked lists, Traversing, Inserting, Deleting and Searching in Singly Linked List, Doubly Linked List and Circular Linked List.<br>**Applications of Linked Lists:** Addition of Polynomials.<br>**Hashing and Collision:** Hashing, Hash Tables, Types of Hash Functions, Collision, Collision Resolution with Open Addressing and Chaining. | 11 |
| III | **Stacks, Queues, and Recursion**<br>**Stacks:** Definition, Representation of Stacks using Arrays and Linked List, Operations on Stacks using Arrays and Linked List.<br>**Application of Stacks:** Arithmetic Expressions, Polish Notation, Conversion of Infix Expression to Postfix Expression, Evaluation of Postfix Expression.<br>**Recursion:** Definition, Recursive Notation, Runtime Stack.<br>**Applications of Recursion:** Factorial of Number, GCD, Fibonacci Series and Towers of Hanoi.<br>**Queues:** Definition, Representation of Queues using Array and Linked List, Types of Queue: Simple Queue, Circular Queue, Double-Ended queue, Priority Queue, Operations on Simple Queues and Circular Queues using Array and Linked List.<br>**Applications of Queues:** Various use cases in problem-solving. | 11 |
| IV | **Graphs and Trees**<br>**Graphs:** Definition, Terminology, Representation, Traversal.<br>**Trees:** Definition, Terminology, Binary Trees, Traversal of Binary Tree, Binary Tree Representation: Array Representation and Pointer (Linked List) Representation.<br>**Binary Search Tree:** Inserting, Deleting and Searching in Binary Search Tree.<br>**Height Balanced Trees:** AVL Trees, Insertion and Deletion in AVL Tree. | 12 |

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| **Lab Programs** | **1. Array Operations:**<br>○ Write a program for insertion and deletion operations in an array.<br>**2. Searching in an Array:**<br>○ Write a program to search for an element in an array using Linear Search and Binary Search.<br>**3. Sorting an Array:**<br>○ Write a program to sort an array using Bubble Sort, Selection Sort, and Insertion Sort.<br>**4. Merging Arrays:**<br>○ Write a program to merge two arrays.<br>**5. Matrix Operations:**<br>○ Write a program to add and subtract two matrices.<br>○ Write a program to multiply two matrices.<br>**6. Singly Linked List Operations:**<br>○ Write a program to insert an element into a Singly Linked List:<br>▪ At the beginning<br>▪ At the end<br>▪ At a specified position<br>○ Write a program to delete an element from a Singly Linked List:<br>▪ At the beginning<br>▪ At the end<br>▪ A specified element<br>**7. Doubly Linked List Operations:**<br>○ Write a program to perform the following operations in a Doubly Linked List:<br>▪ Create<br>▪ Search for an element<br>**8. Circular Linked List Operations:**<br>○ Write a program to perform the following operations in a Circular Linked List:<br>▪ Create<br>▪ Delete an element from the end<br>**9. Stack Operations:**<br>○ Write a program to implement stack operations using an array.<br>○ Write a program to implement stack operations using a linked list.<br>**10. Polynomial Addition:**<br>• Write a program to add two polynomials using linked lists.<br>**11. Postfix Evaluation:**<br>• Write a program to evaluate a postfix expression using a stack.<br>**12. Recursion Operations:**<br>• Write a program to perform the following using recursion:<br>○ Find the factorial of a number<br>○ Find the GCD of two numbers<br>○ Solve the Towers of Hanoi problem<br>**13. Queue Operations:**<br>• Write a program to implement simple queue operations using an array.<br>• Write a program to implement circular queue operations using an array.<br>• Write a program to implement circular queue operations using a linked list.<br>**14. Binary Search Tree (BST) Operations:**<br>• Write a program to perform the following operations on a binary search tree:<br>○ Preorder Traversal<br>○ Inorder Traversal<br>○ Postorder Traversal<br>• Write a program to perform insertion operation in a binary search tree. |